

The Integrity of the Cube is Small

Lowell W. Beineke¹, Wayne Goddard², Peter Hamburger¹,
Daniel J. Kleitman², Marc J. Lipman³ & Raymond E. Pippert¹

Abstract

It is shown that the integrity of the n -dimensional cube is $O(2^n \log n / \sqrt{n})$.

Barefoot, Entringer and Swart [1] introduced the graphical parameter integrity. Let $m(H)$ denote the maximum number of vertices in a component of a graph H . Then the integrity of a graph G , denoted $I(G)$, is defined by $I(G) = \min\{|S| + m(G - S) : S \subset V(G)\}$.

In [2] the values of the integrity of some product graphs were calculated. One such graph for which the integrity was not calculated was the cube. The n -dimensional cube Q_n has 2^n vertices, and may be defined as the cartesian product of n copies of Q_1 , where Q_1 is the complete graph on 2 vertices.

In [2] it was conjectured that $I(Q_n)$ was $2^{n-1} + 1$. We show here that this value is considerably too high.

The Upper Bound

One approach to cutting up the cube is as follows. You find a suitable set $\mathcal{D}_n$ of integers, and then select any vertex x and for each $d \in \mathcal{D}_n$, remove all the vertices of distance d from x . Indeed, this approach can be used to show that $I(Q_n)$ is $O(2^n/n^{1/4})$, and thus that the conjectured value of $I(Q_n)$ is incorrect. But the approach described below is better.

Consider an n -dimensional cube Q_n where n is a power of 2. For any vertex x , we define the *cut* C_x to be the set of vertices at distance $n/2$ from x . We investigate removing a series of “orthogonal” cuts.

¹Department of Mathematical Sciences, Indiana-Purdue University at Fort Wayne, Fort Wayne IN 46805, USA

²Department of Mathematics, Massachusetts Institute of Technology, Cambridge MA 02139, USA

³Office of Naval Research, 800 North Quincy Street, Arlington VA 22217, USA

Choose a collection $x_1, x_2, \dots, x_r$ of vertices which are pairwise distance $n/2$ apart as follows: x_1 is given by the string 0^n , x_2 by $0^{n/2}1^{n/2}$, x_3 by $0^{n/4}1^{n/4}0^{n/4}1^{n/4}$ etc. Note that r is at most $1 + \log n$ (where $\log$ is base 2). Then let $G_r = Q_n - (C_{x_1} \cup C_{x_2} \cup \dots \cup C_{x_r})$.

Lemma. *If $n \geq 4$ then G_r has 2^r isomorphic parts, each the union of components. Thus $m(G_r) \leq 2^{n-r}$.*

PROOF. Label a vertex v in G_r with an r -tuple $(y_1, \dots, y_r)$ where $y_i = +1$ if v is less than $n/2$ away from x_i in Q_n , and -1 otherwise. This labeling partitions G_r into 2^r parts. It is easy to see that there is no edge connecting two parts.

To show that these parts are isomorphic, observe that there exists an automorphism φ_i of Q_n which maps x_i to its complement $\bar{x}_i$ but keeps the other x_j fixed. (For example, consider $n = 4$: the function which maps string x to the reversal of its complement fixes x_2 and x_3 but maps x_1 to 1111.) Thus the composition of the relevant φ_i is the desired isomorphism. QED

Hence we have established that: If n is a power of 2 then

$$I(Q_n) \leq \min_{0 \leq r \leq 1 + \log n} \left\{ 2^{n-r} + r \binom{n}{n/2} \right\}. \quad (1)$$

(It can be shown that the overestimates in this bound are asymptotically insignificant.)

A standard result is that

$$\binom{n}{n/2} \sim \frac{2^n}{\sqrt{n\pi/2}}.$$

Therefore calculus shows that the value of r which minimizes the expression in Equation 1 is approximately $\log(\ln 2 \sqrt{n\pi/2}) + O(\log \log n)$. We choose r to be $\lceil (\log n)/2 \rceil$. Thus there is a constant c such that, when n is a power of 2, $I(Q_n)$ is at most $c2^n \log n / \sqrt{n}$.

Now let n be any positive integer. We consider Q_n as the cartesian product $Q_m \times Q_l$, where m is the largest power of 2 not exceeding n . As a consequence of a general observation in [2], it holds that $I(Q_n) \leq I(Q_m) \times 2^l$. Thus

$$I(Q_n) \leq c2^m \log m / \sqrt{m} \times 2^l < c2^n \log n / \sqrt{n/2},$$

as $m > n/2$.

Hence we may conclude:

Theorem. *The integrity $I(Q_n)$ of the n -dimensional cube Q_n is $O(2^n \log n / \sqrt{n})$.*

We suspect that this might be the right (order of) value. An $\Omega(2^n / \sqrt{n})$ lower bound on the integrity is an immediate consequence of the following result:

Proposition. [3] *If $m(Q_n - S) \leq 2^{n-1}$ then $|S| \geq \binom{n}{\lfloor n/2 \rfloor}$.*

References

- [1] C.A. Barefoot, R. Entringer & H.C. Swart, Vulnerability in graphs—A comparative survey, *J. Combin. Math. Combin. Comput.* **1** (1987), 13–22.
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